



Principles of Finance

Chapter 8

End-of-Chapter Questions

Time Value of Money II: Equal Multiple Payments

Multiple Choice Questions

1. The best example of a constant perpetuity would most likely be _____.

- A. an annuity due
- B. dividends from common stock
- C. preferred stock
- D. an ordinary annuity

Solution: C. As noted in 8.1, the dividend on a preferred stock typically remains constant and is expected to continue indefinitely.

Section 8.1 LO 1 Easy

2. You wish to endow a university chair of accounting for a salary of \$100,000 per year to the recipient. The university will withdraw \$100,000 each year for the recipient's salary. The amount of your gift will remain untouched indefinitely, in perpetuity. The university can lock in a fixed rate for your investment of 2.8% per year. In order to achieve this, what is the approximate amount of the gift you would have to make now?

- A. \$3,103,569
- B. \$3,571,429
- C. \$4,101,218
- D. \$4,227,827

Solution: B. Using the equation $\text{Gift} = \frac{\text{Annual Distribution}}{R_S}$, the amount of the gift should be

$$\frac{\$100,000}{0.028} = \$3,571,429.$$

Section 8.1 LO 2 Easy

3. Preferred stock in Blue Agate Inc. is issued for dividends of \$3.00 per share. The dividends will increase each year at 0.178%, a growing perpetuity. The required rate of return on a stock such as this is 2.5%. At what approximate price will this preferred stock most likely sell today?

- A. \$117.21
- B. \$119.87
- C. \$120.00
- D. \$129.20

Solution: D. The formula for a growing perpetuity is $\text{Price} = \frac{\text{Annual Dividend}}{R_S - G}$; by substitution,

$$\text{Price} = \frac{\$3.00}{0.025 - 0.00178} = \$129.20.$$

Section 8.1 LO 2 Easy

4. Julio's attorney negotiates a structured settlement after an injury, consisting of seven equal payments to Barry of \$150,000 each. The first payment is due today, and the remaining payments will be received in annual amounts, with the second payment occurring one year from now. What is the approximate value of this settlement in today's dollars if Barry uses a discount rate of 5%?
- A. \$911,352
 B. \$867,960
 C. \$746,235
 D. \$1,050,000

Solution: A. Because this is an annuity due, the formula is derived from the present value of an annuity formula and modified for one additional period:

$$\begin{aligned}
 PV_a &= \$150,000 \times \frac{\left[1 - \frac{1}{(1 + 0.05)^7}\right]}{0.05} \times 1.05 \\
 PV_a &= \$150,000 \times \frac{\left(1 - \frac{1}{1.407100}\right)}{0.05} \times 1.05 \\
 PV_a &= \$150,000 \times \frac{1 - 0.710682}{0.05} \times 1.05 \\
 PV_a &= \$150,000 \times \frac{0.289318}{0.05} \times 1.05 \\
 PV_a &= \$150,000 \times 5.78636 \times 1.05 \\
 PV_a &\approx \$911,351.70
 \end{aligned}$$

Section 8.2 LO 4 Medium

5. What is the approximate present value of an ordinary annuity (beginning one year from now) of a stream of 12 annual payments of \$87,000 if you use a discount rate of 6%?
- A. \$773,154.04
 B. \$747,278.92
 C. \$729,394.95
 D. \$718,974.58

Solution: C. Using the present value of an annuity formula:

$$\begin{aligned}
 PV_a &= \$87,000 \times \frac{\left[1 - \frac{1}{(1 + 0.06)^{12}}\right]}{0.06} \\
 PV_a &= \$87,000 \times \frac{\left(1 - \frac{1}{2.012196}\right)}{0.06} \\
 PV_a &= \$87,000 \times \frac{1 - 0.496969}{0.06} \\
 PV_a &= \$87,000 \times \frac{0.503031}{0.06} \\
 PV_a &= \$87,000 \times 8.38385 \\
 PV_a &= \$729,394.95
 \end{aligned}$$

Section 8.2 LO 3 Medium

6. If Maria invests \$2,700 at the end of each six-month period for six years at an annual rate of 4%, what is the approximate future value of his ordinary annuity? Review Chapter 7 for the techniques of interannual compounding.
- A. \$17,909.10
 - B. \$20,248.23
 - C. \$31,755.54
 - D. \$36,212.67

Solution: D. For semiannual compounding, we use the future value of an annuity formula, modifying the rate to 2% and the periods to 12, for two semiannual periods per year:

$$\begin{aligned} FVa &= \$2,700 \times \frac{(1 + 0.02)^{12} - 1}{0.02} \\ FVa &= \$2,700 \times \frac{1.268242 - 1}{0.02} \\ FVa &= \$2,700 \times 13.4121 \\ FVa &= \$36,212.67 \end{aligned}$$

Section 8.2 LO 3 Medium

7. Assume all of the same facts as in exercise 6 above, except that Maria begins immediately and makes each of his payments at the beginning of each 6-month period instead of the end. What is the approximate future value of her annuity due at the end of the six years?
- A. \$17,909.10
 - B. \$36,936.92
 - C. \$32,707.24
 - D. \$22,997.88

Solution: B. The procedure is the same as in question 7 above, except that we modify the ordinary annuity formula to an annuity due by multiplying by one more period:

$$\begin{aligned} FVa &= \$2,700 \times \frac{(1 + 0.02)^{12} - 1}{0.02} \times 1.02 \\ FVa &= \$2,700 \times 13.4121 \times 1.02 \\ FVa &= \$36,936.92 \end{aligned}$$

Section 8.2 LO 3 Medium

8. Rather than spending her \$48,000 in casino winnings, Christy places the money in an investment that will earn her 5% per year, compounded annually. She will withdraw the money in four equal annual installments beginning one year from today. What must the approximate amount of each annual withdrawal be for this investment to be fully depleted in four years?
- A. \$11,136.38
 - B. \$12,892.56
 - C. \$12,243.47
 - D. \$13,536.61

Solution: D. This is an ordinary annuity being solved for the payment amount. We use the present value of an annuity formula, but we already know the present value; we're solving for a different unknown, the payment (PYMT):

$$\begin{aligned}
 \$48,000 &= \text{PYMT} \times \frac{\left[1 - \frac{1}{(1 + 0.05)^4}\right]}{0.05} \\
 \$48,000 &= \text{PYMT} \times \frac{\left(1 - \frac{1}{1.215506}\right)}{0.05} \\
 \$48,000 &= \text{PYMT} \times \frac{0.177297}{0.05} \\
 \$48,000 &= \text{PYMT} \times 3.54594 \\
 \text{PYMT} &= \$48,000 \div 3.54594 \\
 \text{PYMT} &\approx \$13,536.61
 \end{aligned}$$

Section 8.2 LO 3 Medium

9. Your friend Jamal borrows \$5,000 from you, agreeing to pay you back with 8% annual interest, with the first payment due to you one year from today. You ask that you be fully repaid over the next four years. However, to lower his annual payment, Jamal asks you to extend the period over five full years instead. What will be the approximate **difference** in his total payments to you, including interest and principal, if the debt is amortized over five years rather than four?
- A. Jamal will pay \$544 less.
 - B. Jamal will pay \$544 more.
 - C. Jamal will pay \$223 less.
 - D. Jamal will pay \$223 more.

Solution: D. We are comparing two ordinary annuities. While different approaches are possible, here we calculate the payment under each scenario, multiply by the number of years for each, and compare the total of the payments. We will use the formula for the present value of an annuity:

$$\begin{aligned}
 &\text{YOU (4 Payments)} \\
 \$5,000 &= \text{PYMT} \times \frac{\left[1 - \frac{1}{(1 + 0.08)^4}\right]}{0.08} \\
 \$5,000 &= \text{PYMT} \times \frac{\left[1 - \frac{1}{(1.08)^4}\right]}{0.08} \\
 \$5,000 &= \text{PYMT} \times \frac{\left(1 - \frac{1}{1.360489}\right)}{0.08} \\
 \$5,000 &= \text{PYMT} \times \frac{0.264970}{0.08} \\
 \$5,000 &= \text{PYMT} \times 3.312125
 \end{aligned}$$

$$\begin{aligned}
 &\text{JAMAL (5 Payments)} \\
 \$5,000 &= \text{PYMT} \times \frac{\left[1 - \frac{1}{(1 + 0.08)^5}\right]}{0.08} \\
 \$5,000 &= \text{PYMT} \times \frac{\left[1 - \frac{1}{(1.08)^5}\right]}{0.08} \\
 \$5,000 &= \text{PYMT} \times \frac{\left(1 - \frac{1}{1.469328}\right)}{0.08} \\
 \$5,000 &= \text{PYMT} \times \frac{0.319417}{0.08} \\
 \$5,000 &= \text{PYMT} \times 3.992713
 \end{aligned}$$

$$\begin{aligned} \text{PYMT} &= \frac{\$5,000}{3.312125} \\ \text{PYMT} &= \$1,509.60 \end{aligned}$$

$$\begin{aligned} \text{PYMT} &= \frac{\$5,000}{3.992713} \\ \text{PYMT} &= \$1,252.28 \end{aligned}$$

Total Payments:

$$\$1,509.60 \times 4 = \$6,038.40$$

$$\$1,252.28 \times 5 = \$6,261.40$$

Jamal will end up paying $\$6,261.40 - \$6,038.40 = \$223$ more.

Section 8.3 LO 4 Difficult

10. Your new El Supremo credit card arrangement indicates that you will owe interest on unpaid balances at a nominal (stated) rate of 1.2% per month. If the interest rate is compounded monthly, what is the approximate effective annual rate of interest?

- A. 15.39%
- B. 12.00%
- C. 14.02%
- D. 14.40%

Solution: A. Using the formula $(1 + i)^N - 1$ to isolate the effective rate, we substitute:
 $(1 + 0.012)^{12} - 1 = 0.1539 = 15.39\%$.

Section 8.3 LO 4 Easy

Problems

Use four decimal places on time value of money factors unless otherwise specified.

Approximations and minor differences because of rounding are acceptable. Ignore the effect of taxes. Assume that all percentages are annual rates and that compounding occurs annually unless indicated otherwise.

1. Donna enters into an investment contract that will guarantee her 4% per year if she deposits \$3,500 each year for the next 10 years. She must make the first deposit one year from today, the day she signs the agreement. How much will she have when she makes her last payment 10 years from now?

Solution: This is an ordinary annuity with a rate of 4% over 10 years. We solve with the future value of an annuity formula:

$$\begin{aligned} \text{FVa} &= \$3,500 \times \left[\frac{(1 + 0.04)^{10} - 1}{0.04} \right] \\ \text{FVa} &= \$3,500 \times \left(\frac{1.480244 - 1}{0.04} \right) \\ \text{FVa} &= \$3,500 \times \left(\frac{0.480244}{0.04} \right) \\ \text{FVa} &= \$3,500 \times 12.0061 \\ \text{FVa} &= \$42,021.35 \end{aligned}$$

Period	Payment	Interest	Balance
1	3,500.00	-	3,500.00
2	3,500.00	140.00	7,140.00
3	3,500.00	285.60	10,925.60
4	3,500.00	437.02	14,862.62
5	3,500.00	594.50	18,957.13
6	3,500.00	758.29	23,215.41
7	3,500.00	928.62	27,644.03
8	3,500.00	1,105.76	32,249.79
9	3,500.00	1,289.99	37,039.78
10	3,500.00	1,481.59	42,021.37
			(Rounding)

Section 8.2 LO 3 Medium

2. Assume the same facts as in problem 1 above, except that Donna negotiates the chance to make her first payment now and continue to pay at the beginning of each year for the 10-year period. How much will she have accumulated?

Solution: Converting this to an annuity due simply means multiplying the result for an ordinary annuity by one additional period: $\$42,021.35 \times 1.04 = \$43,702.20$. Note that converting an ordinary annuity to an annuity due, whether future value or present value, will always produce a greater result.

Period	Payment	Interest	Balance
1	3,500.00	140.00	3,640.00
2	3,500.00	285.60	7,425.60
3	3,500.00	437.02	11,362.62
4	3,500.00	594.50	15,457.13
5	3,500.00	758.29	19,715.41
6	3,500.00	928.62	24,144.03
7	3,500.00	1,105.76	28,749.79
8	3,500.00	1,289.99	33,539.78
9	3,500.00	1,481.59	38,521.37
10	3,500.00	1,680.85	43,702.23
			(Rounding)

Section 8.2 LO 3 Medium

- Steve purchases preferred stock in Berklee Corporation, with each share paying a \$2.50 dividend. This dividend will remain constant. If the public's required rate of return for Berklee stock is 8%, at what price should this company's stock sell?

Solution: The unchanging dividend on this preferred stock makes it a constant perpetuity. Using the constant perpetuity formula, the stock should sell at $\frac{\$2.50}{0.08} = \31.25 per share.

Section 8.1 LO 2 Easy

- Bill will receive a royalty payment of \$18,000 per year for the next 25 years, beginning one year from now, as a result of a book he has written. If a discount rate of 10 percent is applied, should he be willing to sell out his future rights now for \$160,000? How about \$162,500? \$165,000?

Solution: We discount Bill's expected stream of payments to their present value in order to compare them to his alternative cash offers today, using the present value of an annuity formula:

$$\begin{aligned} PV_a &= \$18,000 \times \frac{\left[1 - \frac{1}{(1 + 0.10)^{25}}\right]}{0.10} \\ PV_a &= \$18,000 \times \frac{\left(1 - \frac{1}{10.834706}\right)}{0.10} \\ PV_a &= \$18,000 \times \frac{0.907704}{0.10} \\ PV_a &= \$18,000 \times 9.07704 \\ PV_a &= \$163,386.72 \end{aligned}$$

All things equal, Bill should not accept a cash offer of less than this amount. He should reject the first two offers but should accept the \$165,000 because it's greater than the present value of the ordinary annuity.

Section 8.2 LO 3 Medium

- Jada is borrowing \$40,000 from you today. She agrees to pay you back in annual installments beginning a year from now over eight years, with interest at 3%. What would her annual payment amount be, including both interest and principal?

Solution: We use the present value of an annuity formula, solving for the payment (PYMT):

$$\begin{aligned} \$40,000 &= PYMT \times \frac{\left[1 - \frac{1}{(1 + 0.03)^8}\right]}{0.03} \\ \$40,000 &= PYMT \times \frac{\left(1 - \frac{1}{1.266770}\right)}{0.03} \\ \$40,000 &= PYMT \times \frac{0.210591}{0.03} \\ \$40,000 &= PYMT \times 7.0197 \\ PYMT &= \frac{\$40,000}{7.0197} \end{aligned}$$

$$\text{PYMT} \approx \$5,698.25$$

Jada's annual payment for interest and principal would be \$5,698.25.

Period	Payment	Interest	Principal	Balance
				40,000.00
1	5,698.25	1,200.00	4,498.25	35,501.75
2	5,698.25	1,065.05	4,633.20	30,868.55
3	5,698.25	926.06	4,772.19	26,096.36
4	5,698.25	782.89	4,915.36	21,181.00
5	5,698.25	635.43	5,062.82	16,118.18
6	5,698.25	483.55	5,214.70	10,903.48
7	5,698.25	327.10	5,371.15	5,532.33
8	5,698.25	165.97	5,532.28	0.05
				(Rounding)

Section 8.3 LO 3 Medium

6. You agree to finance your new SUV with an auto loan of \$38,000. This loan will be repaid over three years with monthly payments (and compounding) at a 4% annual interest rate (0.33% per month). What will your monthly loan payment be?

Solution: Using the present value of annuity formula and solving for the payment, we first adjust 4% for three years to 0.33% for 36 months (periods).

$$\begin{aligned}
 \$38,000 &= \text{PYMT} \times \frac{\left[1 - \frac{1}{(1 + 0.0033)^{36}}\right]}{0.0033} \\
 \$38,000 &= \text{PYMT} \times \frac{\left(1 - \frac{1}{1.125924}\right)}{0.0033} \\
 \$38,000 &= \text{PYMT} \times \frac{0.111841}{0.0033} \\
 \$38,000 &= \text{PYMT} \times 33.891212 \\
 \text{PYMT} &= \frac{\$38,000}{33.891212} \\
 \text{PYMT} &\approx \$1,121.24
 \end{aligned}$$

Your monthly payment will be \$1,121.24.

Section 8.3 LO 3 Medium

7. Debbie won the \$60 million lottery. She is to receive \$1 million a year for the next 50 years beginning one year from now, plus an additional lump sum payment of \$10 million after 50 years. The discount rate is 10 percent. How much cash would she need to be offered today to tempt her to take a lump-sum cash offer instead, all things equal?

Solution: This represents two problems in one, a single payment plus an ordinary annuity. The present value of each must be calculated and then combined to provide Debbie with the total present value of this combination of future payments. Step 1: Calculate the present value of the stream of \$1,000,000 payments using the present value of an annuity formula. Step 2: Calculate the present value of the \$10 million payment she will receive in 50 years using the formula from Time Value of Money I. Step 3: Combine the amounts from Steps 1 and 2 to determine the total present value.

Step 1:

$$\begin{aligned} PV_a &= \$1,000,000 \times \frac{\left[1 - \frac{1}{(1 + 0.10)^{50}}\right]}{0.10} \\ PV_a &= \$1,000,000 \times \frac{\left(1 - \frac{1}{117.390853}\right)}{0.10} \\ PV_a &= \$1,000,000 \times \frac{0.991481}{0.10} \\ PV_a &= \$1,000,000 \times 9.91481 \\ PV_a &= \$9,914,810 \end{aligned}$$

Step 2:

$$\begin{aligned} PV &= FV \times \frac{1}{(1 + i)^n} \\ PV &= \$10,000,000 \times \frac{1}{(1.10)^{50}} \\ PV &= \$10,000,000 \times 0.008519 \\ PV &= \$85,190 \end{aligned}$$

Step 3:

$$\$9,914,810 + \$85,190 = \$10,000,000$$

Debbie should not accept anything less than \$10,000,000 today. Students may be surprised to see that the total result equals this round number; this is coincidental.

Section 8.2 LO 3 Difficult

8. Kim started a paper route on January 1, 2016. Every three months, she deposited \$300 in her new bank account, which earned 4 percent annually but was compounded quarterly. On December 31, 2019, she placed the entire balance in her bank account in an investment that earned 5 percent annually. How much will she have on December 31, 2022?

Solution: This is a two-stage future value problem combining an ordinary annuity with a single amount. Step 1: Calculate the future value of an annuity that is increased and compounded quarterly for four years. Step 2: Calculate the future value of a single amount that is compounded annually for three years, beginning with the result from Step 1.

Step 1:

$$\begin{aligned}FVa &= \$300 \times \frac{(1 + 0.01)^{16} - 1}{0.01} \\FVa &= \$300 \times \frac{0.172579}{0.01} \\FVa &= \$300 \times 17.2579 \\FVa &= \$5,177.37\end{aligned}$$

Step 2:

$$\begin{aligned}FV &= PV \times (1 + i)^n \\FV &= \$5,177.37 \times (1 + 0.05)^3 \\FV &= \$5,177.37 \times 1.157625 \\FV &= \$5,993.45\end{aligned}$$

Kim will have \$5,993.45 at the end of the seven years. It is common for students to overlook the fact that the annuity lasts for four years, not three (1/1/2016 through 12/31/2019).

Section 8.2 LO 3 Difficult

9. You hire Thomas to work for you for five years, and you agree to put away enough money as a lump sum now to fund an annuity for him. At the end of those five years, he will retire and may begin drawing out \$20,000 per year for five years, starting on the last day of each year (in this case, the end of year 6, from when this arrangement began, through year 10). How much must you invest today if your guaranteed interest rate is 3% compounded annually for all 10 years?

Solution: Another two-stage present value problem, involving first finding a present value at a starting point (even though it occurs in our future!) that will generate a series of future payments and then calculating a single-amount present value today to achieve that future goal when payments (withdrawals) will begin. This is a challenging problem. At a point five years in the future, we will need an amount to fund a flow of five \$20,000 payments, an ordinary annuity, at 3%. Students are encouraged to use a timeline in analyzing this problem.

Step 1: Determine the amount needed in five years to begin the ordinary annuity that must then last five years, using the present value of an annuity formula. The fact that this is occurring five years in the future is irrelevant. Step 2: Using that amount, discount it at a single amount to today's value using the present value of a single amount formula.

Step 1:

$$\begin{aligned}
 PVa &= \$20,000 \times \frac{\left[1 - \frac{1}{(1 + 0.03)^5}\right]}{0.03} \\
 PVa &= \$20,000 \times \frac{\left(1 - \frac{1}{1.159274}\right)}{0.03} \\
 PVa &= \$20,000 \times \frac{0.137391}{0.03} \\
 PVa &= \$20,000 \times 4.5797 \\
 PVa &= \$91,594
 \end{aligned}$$

Step 2:

$$\begin{aligned}
 PV &= FV \times \frac{1}{(1 + i)^n} \\
 PV &= \$91,594 \times \frac{1}{(1 + 0.03)^5} \\
 PV &= \$91,594 \times \frac{1}{1.159274} \\
 PV &= \$91,594 \times 0.862609 \\
 PV &= \$79,009.81
 \end{aligned}$$

If you invest \$79,009.81 today and it grows at 3% per year for five years, it will then total \$91,594. At that point, an ordinary annuity begins, so the fund grows for one more year before Thomas receives his first payment of \$20,000. This process continues for four more years, and when he receives his fifth and final payment, the fund will be fully depleted (allowing for minor rounding differences). A proof follows:

Step 1	Period	NA	Interest	Balance
				79,008.98
	1		2,370.27	81,379.25
	2		2,441.38	83,820.63
	3		2,514.62	86,335.25
	4		2,590.06	88,925.30
	5		2,667.76	91,593.06
				(Rounding)
Step 2	Period	Payment	Interest	Balance
				91,594.00
	1	20,000.00	2,747.82	74,341.82
	2	20,000.00	2,230.25	56,572.07
	3	20,000.00	1,697.16	38,269.24
	4	20,000.00	1,148.08	19,417.31
	5	20,000.00	582.52	(0.17)
				(Rounding)

Section 8.2 LO 3 Difficult

10. Your new boss doesn't have a pension or 401(k) plan for your retirement, but she agrees to place aside \$12,000 every year once a year for four years. She gives you the option of either starting immediately on your first day of work or starting one year from now. That makes this the difference between an ordinary annuity and an annuity due. If the plan earns 5% per year, compounded annually, what will be the difference between the two approaches be after the four years / four payments?

Solution: We solve for both the future value of an ordinary annuity and the future value of an annuity due. Taking the two future values and subtracting them provides the excess of the annuity due over the ordinary annuity; at this point, we eliminate the reduction, having completed several practice problems above:

$$\text{Ordinary Annuity: FVa} = \$12,000 \times \frac{(1 + 0.05)^4 - 1}{0.05}$$

$$\text{Ordinary Annuity: FVa} = \$51,721.50$$

$$\text{Annuity Due: FVa} = \$12,000 \times \frac{(1 + 0.05)^4 - 1}{0.05} \times 1.05$$

$$\text{Annuity Due: FVa} = \$54,307.58$$

The annuity due would exceed the ordinary annuity by

$$\$54,307.58 - \$51,721.50 = \$2,586.08$$

Section 8.2 LO 3 Difficult

Chapter 8 Test Bank

Time Value of Money II: Equal Multiple Payments

These questions and problems can all be solved in a classroom setting using a handheld financial calculator.

True or False Questions

1. The present value of an annuity due will higher be than that of an ordinary annuity.

- A. True
- B. False

Solution: A. True

Section 8.2 LO 3 Easy

2. Amortizing a loan via equal payments will result in the early payments being composed more of interest than principal repayments.

- A. True
- B. False

Solution: A. True

Section 8.3 LO 3 Easy

3. If correctly calculated, a loan amortization schedule will bring the balance remaining due on the loan to zero upon reaching its due date.

- A. True
- B. False

Solution: A. True

Section 8.3 LO 3 Easy

Multiple Choice Questions

1. What is the present value of a constant perpetuity of \$17 per year if the discount rate is 10 percent?

- A. \$188
- B. \$170
- C. \$165
- D. \$190

Solution: B. Apply the formula PMT/r where $PMT = 17$, $r = 10$ percent

Section 8.1 LO 2 Easy

2. What is the present value of a perpetuity of \$17 per year growing at 2 percent per year with a discount rate of 10 percent?

A. \$234
B. \$200
C. \$150
D. \$320

Solution: A. Apply the formula $PMT \times (1 + g)/(r - g)$ where $PMT = 17$, $g = 2$ percent, $r = 10$ percent

Section 8.2 LO 1 Easy

3. What is the present value of a \$1,380 annuity due making payments for the next 5 years at an interest rate of 3 percent?

A. \$6,319.99
B. \$6,509.59
C. \$6,402.20
D. \$5,960.30

Solution: B. Solve for the PV of an annuity due, given $PMT = 1,380$, $n = 5$, $r = 3$ percent

Section 8.2 LO 2 Moderate

4. What is the present value of annual payments of \$2,000 for 12 years at 1 percent?

A. \$24,000
B. \$22,510
C. \$19,740
D. \$25,140

Solution: B. Solve for PV, given $PMT = 2,000$, $n = 12$, $r = 1$ percent

Section 8.2 LO 3 Easy

5. You win a lottery payout of \$10,000 a year for the next 30 years. If your investment rate is 8 percent, how much would an acceptable lump sum payment be?

A. \$112,578
B. \$203,010
C. \$300,000
D. \$145,309

Solution: A. Solve for PV, given $PMT = 10,000$, $n = 30$, $r = 8$ percent

Section 8.2 LO 4 Moderate

6. If you have \$750,000 and invest money at 6 percent, how much can you withdraw annually over the next 30 years?
- A. \$25,000
 - B. \$45,010
 - C. \$54,486
 - D. \$65,302

Solution: C. Solve for PMT, given $n = 30$, $r = 6$ percent, $PV = 750,000$

Section 8.2 LO 5 Moderate

7. At what rate must you invest \$2,500 per year if you want to have \$12,000 in your investment account in 8 years?
- A. 10.75 percent
 - B. 11.75 percent
 - C. 12.99 percent
 - D. 13.52 percent

Solution: C. Solve for r , given $PMT = 2,500$, $n = 8$, $FV = 12,000$

Section 8.2 LO 5 Moderate

8. How long will it take to have \$18,000 in your account if you invest \$1,250 per year at 3 percent interest?
- A. 12 years
 - B. 13 years
 - C. 17 years
 - D. 15 years

Solution: B. Solve for n , given $FV = 18,000$, $PMT = 1,250$, $r = 3$ percent

Section 8.2 LO 5 Moderate

9. If you save \$10,000 a year at 7 percent interest, how much will you accumulate in 30 years?
- A. \$944,608
 - B. \$300,000
 - C. \$875,034
 - D. \$1,020,300

Solution: A. Solve for FV , given $PMT = 10,000$, $r = 7$ percent, $n = 30$

Section 8.2 LO 5 Moderate

10. You are given a choice of investing at 2.9 percent compounded semiannually or 2.85 percent compounded quarterly. Which of these choices will give you a better rate of return?

- A. Compounding semiannually
- B. Compounding quarterly
- C. Cannot be determined from the information
- D. Same rate for both

Solution: A. Solve for EAR, given r and n where $r = .029$ and $n = 2$ for the semiannual option and $r = 0.285$ and $n = 4$ for the quarterly option

Section 8.4 LO 1 Moderate

11. The effective annual rate of 3 percent compounded semiannually is _____.

- A. 3.0225 percent
- B. 4.1020 percent
- C. 2.9051 percent
- D. 3.2030 percent

Solution: A. Solve for EAR, given $r = .03$ and $n = 2$

Section 8.4 LO 1 Moderate

12. How much more will \$1,230 be worth at the end of 5 years if it is invested at 5 percent compounded semiannually as compared to being invested at 5 percent annually? Note that the question is asking for the incremental amount above that of the result of the annually compounded amount.

- A. \$5.20
- B. \$4.67
- C. \$3.96
- D. \$8.02

Solution: B. Solve for FV, given $PV = 1,230$, $n = 5$, $r = .05$ and for FV, given $PV = 1,230$, $n = 10$, $r = 0.25$. Then compare results.

Section 8.4 LO 1 Moderate

13. What is the future value of annual payments of \$3,000 for 15 years at 2 percent?

- A. \$45,000
- B. \$62,010
- C. \$51,880
- D. \$85,300

Solution: C. Solve for FV, given $PMT = 3,000$, $n = 15$, $r = 2$ percent

Section 8.5 LO 2 Easy

14. How much must you pay annually to amortize a \$30,000 car loan at 8 percent over the next 5 years?

- A. \$6,000
- B. \$8,250
- C. \$6,975
- D. \$7,514

Solution: D. Solve for PMT, given $PV = 30,000$, $n = 5$, $r = 8$ percent

Section 8.5 LO 4 Moderate

Problems

1. Solve for the future value of \$1,300 after 1 year invested at 4 percent if compounding is a) annual, b) semiannual, c) quarterly, and d) monthly.

Solution: Solve for FV, given $PV = 1,300$ and using $r = .04$ divided by 1, 2, 4, and 12 successively.

a) 1,352, b) 1,352.52, c) 1,352.78, d) 1,352.96

Section 8.4 LO 1 Moderate

2. Solve for the present value of \$21,500 one year into the future invested at 3 percent if compounding is a) annual, b) semiannual, c) quarterly, and d) monthly.

Solution: Solve for PV, given $FV = 21,500$ and using $r = .03$ divided by 1, 2, 4, and 12 successively.

a) 2,575, b) 2,575.56, c) 2,575.85, d) 2,576.04

Section 8.4 LO 1 Moderate

3. What is the effective annual rate of 3 percent compounded a) annually, b) semiannually, c) quarterly, and d) monthly?

Solution: Solve for EAR, given $r = .03$ and varying n among 1, 2, 4, and 12. a) .03, b) .03225, c) .03339, d) .030416

Section 8.4 LO 1 Moderate

4. If you want to convert \$10,000 today into an annual annuity payment over the next 5 years, what would that payment be at a) 3 percent interest, b) 4 percent interest, and c) 5 percent interest?

Solution: Solve for PMT, given $PV = 10,000$, $n = 5$ and varying r from 3 percent to 5 percent. a) 2,183.55, b) 2,246.27, c) 2,309.75

Section 8.5 LO 2 Moderate

5. What would the present value be of an annual annuity payment of \$2,500 over 7 years at an interest rate of a) 6 percent, b) 7 percent, and c) 8 percent?

Solution: Solve for PV, given $PMT = 2,500$, $n = 7$ and varying r from 6 percent to 8 percent. a) 13,956, b) 13,473, c) 13,016

Section 8.5 LO 2 Moderate

6. What is the future value of annual annuity payments of \$1,125 over the next 12 years at interest rates of a) 2 percent, b) 3 percent, and c) 4 percent?

Solution: Solve for FV, given $PMT = 1,125$, $n = 12$ and varying r from 2 percent to 4 percent. a) 15,088, b) 15,966, c) 16,904

Section 8.5 LO 2 Moderate

7. What interest rate must you invest your annual annuity payment of \$3,040 for the next 8 years in order to generate a future value of a) \$30,400, b) \$60,800, and c) \$91,200?

Solution: Solve for r , given $PMT = 3,040$ and varying FV among 30,400, 60,800, and 91,200. a) 6.28 percent, b) 25.21 percent, c) 36.18 percent

Section 8.5 LO 2 Moderate